

Topic: Relationship Types and Relationship Sets

Q. Explain the concepts of Relationship Types and Relationship Sets. Discuss the Degree of Relationship and Mapping Cardinalities (1:1, 1:N, M:N) with examples.

📌 Definition to Remember

A **Relationship Type** defines a meaningful association between two or more entity types, represented by a **Diamond** in ER diagrams. A **Relationship Set** is the actual collection of specific relationship instances existing in the database at any given moment.

1. Relationship Type and Relationship Set

- **Relationship Type:** Association between entity types. E.g., `WORKS_FOR` links `EMPLOYEE` and `DEPARTMENT`.
- **Relationship Set:** All specific instances of that relationship. E.g., "John works for IT Dept." is one instance.

2. Degree of Relationship

The **Degree** = number of entity types participating in the relationship.

Degree	Name	Example
1	Unary	Employee <code>Manages</code> another Employee
2	Binary	Student <code>Enrolls_In</code> Course (most common)
3	Ternary	Doctor, Patient, Drug connected by <code>Prescribes</code>

3. Mapping Cardinalities

Mapping Cardinality defines the maximum number of relationship instances an entity can participate in.

1. One-to-One (1:1)

- One entity in A associates with **at most one** in B, and vice versa.
- **Example:** One `MANAGER` manages one `DEPARTMENT`; one `DEPARTMENT` has one `MANAGER`.

2. One-to-Many (1:N)

- One entity in A associates with **multiple** entities in B; but each B associates with **one** A.
- **Example:** One `DEPARTMENT` has many `EMPLOYEE`s; each `EMPLOYEE` belongs to one `DEPARTMENT`.

3. Many-to-One (N:1)

- Multiple entities in A associate with **one** entity in B.
- **Example:** Many `STUDENT`s belong to one `COURSE`.

4. Many-to-Many (M:N)

- Entities in A can associate with **multiple** in B, and vice versa.
- **Example:** A `STUDENT` enrolls in many `COURSE`s; a `COURSE` has many `STUDENT`s.

1:1 → A ——— <R> ——— B
 1:N → A ═══ <R> ——— B
 M:N → A ═══<R>═══ B

4. Participation Constraints

Type	Description	ER Symbol
Total Participation	Every entity in the set MUST participate	Double Line (══)
Partial Participation	Entity may or may not participate	Single Line (—)

- **Total Example:** Every `EMPLOYEE` must belong to a `DEPARTMENT` .
- **Partial Example:** A `STUDENT` may or may not live in a `HOSTEL` .

★ Must-Write Points (for 10 marks)

1. Relationship Type = association between entity types (Diamond); Relationship Set = collection of all instances.
2. Degree: Unary (1), Binary (2 — most common), Ternary (3).
3. Mapping Cardinalities: 1:1, 1:N, N:1, M:N — define how many entities can participate.
4. 1:1: one entity maps to exactly one (e.g., Manager → Department).
5. 1:N: one on one side, many on the other (e.g., Department → Employees).
6. M:N: many on both sides — requires a Junction Table in the relational model.
7. Total Participation (double line): every entity **MUST** participate; Partial (single line): optional.

⚡ Quick Recall

Relationship Type (Diamond) → Degree (Unary/Binary/Ternary) → Cardinality (1:1, 1:N, M:N)
 → Participation (Total=double line, Partial=single line)